Course: Linear Algebra and Optimization **Course Code:** 23MA4BSLAO

Unit – 1: Continuous Optimisation-1

Gradient of scalar with respect to vector

Let $\phi = \phi(x_1, x_2, \dots, x_n)$ be the scalar function and $x = [x_1 \quad x_2 \quad \cdots \quad x_n]$ be the row vector then gradient of ϕ w.r.t. x is $\frac{d\phi}{dx} = \begin{bmatrix} \frac{d\phi}{dx_1} & \frac{d\phi}{dx_2} & \cdots & \frac{d\phi}{dx_n} \end{bmatrix}$. (Numerator Layout)

Let $\phi = \phi(x_1, x_2, \dots, x_n)$ be the scalar function and $\mathrm{X} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ be the column vector then

gradient of ϕ w.r.t. X is $\frac{d\phi}{d\mathrm{X}} = \begin{bmatrix} \frac{d\phi}{dx_1} \\ \frac{d\phi}{dx_2} \\ \vdots \\ \frac{d\phi}{dx_n} \end{bmatrix}$. (Denominator Layout)

Functions of several variables and partial differentiation

Examples:

Find the gradient of following functions at the given point:

1. $u = e^{xyz}$ and $p = (1,1,1)$	4. $g = 3x^2 + y^2 + z^2$ and $p = (0,0,2)$
2. $f = xyz$ and $p = (1,0,1)$	5. $g = 3xe^{2y} + e^z y^2 + z^2$ and $p = (1,0,0)$
3. $g = x^2 y^2 z$ and $p = (2,3,-1)$	6. $f = 2x^3 e^{y^2} z^2$ and $p = (2,0,3)$

Jacobian Matrix

The **Jacobian matrix** is a matrix composed of the first-order partial derivatives of a multivariable function. Let $\boldsymbol{f}: \mathbb{R}^{\boldsymbol{n}} \to \mathbb{R}^{\boldsymbol{m}}$ be defined by $\boldsymbol{f}(x_1, x_2, \dots, x_n) = (f_1, f_2, \dots, f_m)$ then Jacobian of f is defined as

$$J_X(\boldsymbol{f}) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \frac{\partial f_m}{\partial x_2} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix} = \begin{bmatrix} \frac{df_1}{d\boldsymbol{x}} \\ \frac{df_2}{d\boldsymbol{x}} \\ \vdots \\ \frac{df_n}{d\boldsymbol{x}} \end{bmatrix} = \frac{d\boldsymbol{f}}{d\boldsymbol{X}} = \nabla \boldsymbol{f}$$

Examples:

Find the Jacobian matrix of the following function:

1. $f(x, y) = (x^4 + 3y^2 x, 5y^2 - 2xy + 1)$ at $(1,2)$.
2. $f(x, y) = (e^{xy} + y, y^2 x)$ at $(0, -2)$.
3. $f(x, y) = (x^3 y^2 - 5x^2 y^2, y^6 - 3y^3 x + 7)$ at $(1, -1)$.

4. $f(x,y,z) = \left(z\tan(x^2 - y^2),\ \ xy\ln\left(\frac{z}{2}\right)\right)$ at $(2,-2,2)$.
5. $f(x,y,z) = \left(xe^{2y}\cos(-z),\ \ (y-2)^3\sin\left(\frac{z}{2}\right),\ e^{2y}\ln\left(\frac{x}{3}\right)\right)$ at $(3,0,\pi)$.
6. $f(x,y) = \left(\frac{\cos(x-y)}{x}, e^{x^2-y^2}, x^3\sin(2y)\right)$ at $(1,1,0)$.

Local and Global Optima

Definition: Let $X \subset R^2$, let $f: R^2 \to R$ be a function, and let $p \in X$.

1. f has a local maximum at P if there exist neighbourhood U of P in X such that $f(P) \geq f(X)$ for all $X \in U$.
2. f has a local minimum at P if there exist neighbourhood U of P in X such that $f(P) \leq f(X)$ for all $X \in U$.
3. f has a global maximum at P if $f(P) \geq f(X)$ for all $X \in U$, in which case we call $f(p)$ the global maximum value of f.
4. f has a global minimum at P if $f(P) \leq f(X)$ for all $X \in U$, in which case we call $f(p)$ the global minimum value of f.

Second derivative test: Let $z = f(x,y)$ be the differentiable function. For a critical point (a,b), let

$$D = f_{xx}(a,b)f_{yy}(a,b) - \left(f_{xy}(a,b)\right)^2$$

be the discriminant.

- If $D > 0$ and $f_{xx}(a,b) > 0$, then f has a local minimum at (a,b).
- If $D > 0$ and $f_{xx}(a,b) < 0$, then f has a local maximum at (a,b).
- If $D < 0$, then (a,b) is the saddle point of f.
- If $D = 0$, then the test gives no information.

The Taylor series is

$$f(X) = f(X_0) + \nabla f_0^T(X - X_0) + \frac{1}{2}(X - X_0)^T H_0 (X - X_0) + \cdots$$

To find optimum point of multivariate functions more than two variables, we need to compute the Hessian matrix at all the critical points. In the 2-D case the determinant of the Hessian Matrix is D.

Hessian matrix of function of three variable $f(x,y,z)$ is given by $H = \begin{bmatrix} f_{xx} & f_{xy} & f_{xz} \\ f_{yx} & f_{yy} & f_{yz} \\ f_{zx} & f_{zy} & f_{zz} \end{bmatrix}$

Second derivative test: Find the critical points, plug them in the Hessian matrix, and compute their eigenvalues.

- If all the eigenvalues are strictly positive, then the critical point is a local minimum.
- If all the eigenvalues are strictly negative, then the critical point is a local maximum.
- If H has both positive and negative eigenvalues and all are non-zero, then the critical point is saddle point.
- If any of the eigenvalue is zero, then the test gives no information.

Examples:

Find all the stationary points of the function, the Hessian matrix and hence classify the stationary points for the following multivariate function.

1. $f(x,y) = x^2 + y^2 - 4x + 2y$	5. $f(x,y,z) = x^2 + y^2 + z^2$
2. $f(x,y,z) = xy + yz + zx - 4x + 2y$	6. $f(x,y,z) = 2x^2 + 3y^2 + 4z^2 - 4xyz$
3. $f(x,y,z) = x^2 + y^2 + z^2 + xy + yz + zx$	7. $f(x,y,z) = 3x^2 + 2y^2 + 5z^2 - 6xyz$
4. $f(x,y,z) = -x^3 + 3xz + 2y - y^2 - 3z^2$	8. $f(x,y,z) = x^3 + y^2 + z^2 - 3xyz$

Gradient of vector with respect to vector

If $\boldsymbol{f}: \mathbb{R}^{\boldsymbol{n}} \to \mathbb{R}^{\boldsymbol{m}}$ then the gradient of $\boldsymbol{f} = (f_1, f_2 \dots, f_{\boldsymbol{m}})$ with respect to $\boldsymbol{X} = (x_1, x_2, x_3 \dots, x_n)$ is

$$\frac{d\boldsymbol{f}}{d\mathbf{X}} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \frac{\partial f_n}{\partial x_2} & \cdots & \frac{\partial f_n}{\partial x_n} \end{bmatrix} = \begin{bmatrix} \frac{df_1}{d\boldsymbol{X}} \\ \frac{df_2}{d\boldsymbol{X}} \\ \vdots \\ \frac{df_n}{d\boldsymbol{X}} \end{bmatrix} = \boldsymbol{J}_{\boldsymbol{X}}(\boldsymbol{f})$$

Gradient of matrix with respect to vector

Let $\boldsymbol{f} = \begin{bmatrix} f_{11} & f_{12} & \cdots & f_{1n} \\ f_{21} & f_{22} & \cdots & f_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ f_{m1} & f_{m2} & \cdots & f_{mn} \end{bmatrix}$ and $\boldsymbol{X} = [x_1 \quad x_2 \quad \cdots \quad x_n]$ then gradient of f w.r.t. $\boldsymbol{X}$ is

$$\frac{d\boldsymbol{f}}{d\mathbf{X}} = \begin{bmatrix} \frac{df_{11}}{d\mathrm{X}} & \frac{df_{12}}{d\mathrm{X}} & \cdots & \frac{df_{1n}}{d\mathrm{X}} \\ \frac{df_{21}}{d\mathrm{X}} & \frac{df_{22}}{d\mathrm{X}} & \cdots & \frac{df_{2n}}{d\mathrm{X}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{df_{m1}}{d\mathrm{X}} & \frac{df_{m2}}{d\mathrm{X}} & \cdots & \frac{df_{mn}}{d\mathrm{X}} \end{bmatrix}$$

Note: Each element in the above matrix is of size $1 \times n$.

Gradient of matrix with respect to matrix

Let $f = \begin{bmatrix} f_{11} & f_{12} & \cdots & f_{1n} \\ f_{21} & f_{22} & \cdots & f_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ f_{m1} & f_{m2} & \cdots & f_{mn} \end{bmatrix}$ and $\mathrm{X} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \cdots & x_{mn} \end{bmatrix}$ then gradient of f w.r.t. X is

$$\frac{df}{d\mathrm{X}} = \begin{bmatrix} \frac{df_{11}}{d\mathrm{X}} & \frac{df_{12}}{d\mathrm{X}} & \cdots & \frac{df_{1n}}{d\mathrm{X}} \\ \frac{df_{21}}{d\mathrm{X}} & \frac{df_{22}}{d\mathrm{X}} & \cdots & \frac{df_{2n}}{d\mathrm{X}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{df_{m1}}{d\mathrm{X}} & \frac{df_{m2}}{d\mathrm{X}} & \cdots & \frac{df_{mn}}{d\mathrm{X}} \end{bmatrix}$$

Note: Each element in the above matrix is of size $m \times n$.

Useful identities for computing Gradient

1. $\frac{\partial\left(\mathrm{Trace}(\boldsymbol{f}(\boldsymbol{X}))\right)}{\partial \boldsymbol{X}} = \mathrm{Trace}\left(\frac{\partial \boldsymbol{f}(\boldsymbol{X})}{\partial \boldsymbol{X}}\right)$ 2. $\frac{\partial(\boldsymbol{f}(\boldsymbol{X})^T)}{\partial \boldsymbol{X}} = \left(\frac{\partial \boldsymbol{f}(\boldsymbol{X})}{\partial \boldsymbol{X}}\right)^T$ 3. $\frac{\partial\left(\det(\boldsymbol{f}(\boldsymbol{X}))\right)}{\partial \boldsymbol{X}} = \det(\boldsymbol{f}(\boldsymbol{X}))\mathrm{Trace}\left(\left(f(X)\right)^{-1}\left(\frac{\partial(f(X))}{\partial X}\right)\right)$	4. $\frac{\partial a^T X^{-1} b}{\partial X} = -(X^{-1})^T a b^T (X^{-1})^T$ 5. $\frac{\partial x^T a}{\partial x} = a^T = \frac{\partial a^T x}{\partial x}$ 6. $\frac{\partial a^T X b}{\partial X} = a b^T$

Examples:

1. Discuss the gradient of vector with respect to matrix.
2. Obtain the Gradient of vector $\boldsymbol{f} = \begin{bmatrix} e^{x_0 x_1} & e^{x_2 x_3} \end{bmatrix}$ with respect to the matrix $\mathbf{x} = \begin{bmatrix} x_0 & x_1 \\ x_2 & x_3 \end{bmatrix}$.
3. Obtain the gradient of scalar $\phi = 4x_0 + 2x_1 - 3x_2 + x_4$ with respect to the matrix $\vec{x} = \begin{bmatrix} x_0 & x_1 \\ x_2 & x_3 \end{bmatrix}$.
4. Find the derivative of the vector $\begin{bmatrix} x_0 x_1 \\ x_0 + x_1 \\ x_2 \end{bmatrix}$ with respect to vector $\begin{bmatrix} x_0 & x_1 & x_2 \end{bmatrix}$ and hence find the nature of the resulting matrix at [0,1,1].
5. Obtain the gradient of the vector $\begin{bmatrix} 4x_0 x_1 \\ 4x_0 + 2x_1 \\ 3x_2 \end{bmatrix}$ with respect to vector $\begin{bmatrix} x_0 & x_1 & x_2 \end{bmatrix}$ and hence find the values of x_0 and x_1 for which 8, 3 and 2 are the eigenvalues of derivative matrix.
6. Obtain the gradient of the matrix $\begin{bmatrix} x_0 x_1 \\ x_0 + e^{x_1} \\ x_2^2 \end{bmatrix}$ with respect to $\begin{bmatrix} x_0 \\ x_1 \\ x_2 \end{bmatrix}$.
7. Obtain the gradient of matrix $\mathbf{f} = \begin{bmatrix} \sin(x_0 + 2x_1) & 2x_1 + x_3 \\ 2x_0 + x_2 & \cos(2x_2 + x_3) \end{bmatrix}$ with respect to the matrix $\mathbf{x} = \begin{bmatrix} x_0 & x_1 \\ x_2 & x_3 \end{bmatrix}$.
8. Find the derivative of the matrix $\mathbf{f} = \begin{bmatrix} x_0^2 x_1 x_2 & x_1^2 x_2 x_3 \\ x_3^2 x_1 x_3 & x_2^3 x_1 \end{bmatrix}$ with respect to the matrix $\mathbf{x} = \begin{bmatrix} x_0 & x_1 \\ x_2 & x_3 \end{bmatrix}$.
9. Obtain the gradient of matrix $\begin{bmatrix} xyz & xy \\ zy & zx \\ z & 2yx \end{bmatrix}$ with respect to $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$.
10. Obtain the gradient of the matrix $\begin{bmatrix} xy & yz & zw \\ wx & wy & wz \end{bmatrix}$ with respect to $\begin{bmatrix} x & y \\ z & w \end{bmatrix}$.

Verification problems on gradient identities

1. Given $\mathbf{f} = \begin{bmatrix} x_0^2 x_1 x_2 & x_1^2 x_2 x_3 \\ x_3^2 x_1 x_3 & x_2^3 x_1 \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} x_0 & x_1 \\ x_2 & x_3 \end{bmatrix}$. Verify
   i. $\frac{\partial\left(\mathrm{Trace}(f(X))\right)}{\partial X} = \mathrm{Trace}\left(\frac{\partial f(X)}{\partial X}\right)$
   ii. $\frac{\partial (f(X)^T}{\partial X} = \left(\frac{\partial f(X)}{\partial X}\right)^T$
2. Given $\mathbf{f} = \begin{bmatrix} \sin(x_0 + 2x_1) & 2x_1 + x_3 \\ 2x_0 + x_2 & \cos(2x_2 + x_3) \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} x_0 & x_1 \\ x_2 & x_3 \end{bmatrix}$. Verify
   i. $\frac{\partial\left(\mathrm{Trace}(f(X))\right)}{\partial X} = \mathrm{Trace}\left(\frac{\partial f(X)}{\partial X}\right)$
   ii. $\frac{\partial (f(X)^T}{\partial X} = \left(\frac{\partial f(X)}{\partial X}\right)^T$
3. Given $X = \begin{bmatrix} x_1 & x_2 \\ x_3 & x_4 \end{bmatrix}$, $a = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ and $b = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$. Verify $\frac{\partial (a^T X b)}{\partial X} = ab^T$.
4. Given $X = \begin{bmatrix} x_1 & x_2 \\ x_3 & x_4 \end{bmatrix}$, $a = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ and $b = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. Verify $\frac{\partial (a^T X b)}{\partial X} = ab^T$.
5. Given $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ and $a = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}$. Verify $\frac{\partial (x^T a)}{\partial x} = a^T$.

6. Given $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$ and $a = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 1 \end{bmatrix}$. Verify $\frac{\partial(a^T x)}{\partial x} = a^T$.

Convex sets and functions of separating hyperplanes

Convex set definition: A set C is convex if the line segment between any two points in C lies in C. That is, a set $C \subseteq \mathbb{R}^n$ is convex, if for all $x, y \in C$ and $\forall \lambda \in [0,1]$, $\lambda x + (1 - \lambda) y \in C$.

A point of the form $\lambda x + (1 - \lambda) y$, $\lambda \in [0, 1]$ is called a convex combination of x and y. As λ varies between $[0, 1]$, a line segment is being formed between x and y.

Examples:

1. Show that intersection of two convex sets is a convex set. Is union of two convex sets is necessarily convex set. Justify your answer.
2. Show that the unit ball $B = \{u \in R^n : ||u|| < 1\}$ is convex set.
3. Show that $S = \{(x_1, x_2) : 2x_1 + 3x_2 = 7\} \in R^2$ is a convex set.
4. Show that $S = \{(x_1, x_2) : x_1^2 + x_2^2 \leq 4\}$ is a convex set.
5. Show that $S = \{(x_1, x_2, x_3) : x_1^2 + x_2^2 + x_3^2 \leq 1\}$ is a convex set.
6. Show that $S = \{(x_1, x_2, x_3) : 2x_1 - x_2 + x_3 \leq 4\}$ is a convex set.

Hyperplane

Hyperplane definition: For given scalar d in $\mathbb{R}$, the set of all $(x_1, x_2, \dots x_n) \in R^n$ such that $a_1 x_1 + a_2 x_2 + \cdots + a_n x_n = d$ $(f(X) = d)$ is called hyperplane. That is, if $\boldsymbol{n} = (\boldsymbol{a_1}, \boldsymbol{a_2}, \dots \boldsymbol{a_n})$ and d is scalar then hyperplane is the set $\mathrm{H} = \{X \in R^n : \boldsymbol{n} \cdot X = d\}$.

Note: The vector $\boldsymbol{n}$ is normal to the hyperplane. Analyse How?

Definition: The hyperplane $\mathrm{H} = [f : d]$ separates two sets A and B if one of the following holds:

i. $f(A) \leq d$ and $f(B) \geq d$, or
ii. $f(A) \geq d$ and $f(B) \leq d$.

In the conditions above, if all the weak inequalities are replaced by strict inequalities, then H is said to strictly separate A and B.

Examples:

1. Let $A = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ and $v = \begin{bmatrix} 1 \\ -6 \end{bmatrix}$, and let $\mathrm{H} = \{\mathrm{X} : \mathrm{A} \cdot \mathrm{X} = 12\}$. Find the implicit description of the parallel hyperplane $\mathrm{H}_1 = \mathrm{H} + V$.
2. Let $v_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 4 \\ 0 \end{bmatrix}$. Find an implicit description of hyperplane H that passes through v_1 and v_2.
3. Let $v_1 = \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}$, $v_2 = \begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix}$ and $v_3 = \begin{bmatrix} -1 \\ -2 \\ 5 \end{bmatrix}$. Find an implicit description of hyperplane H that passes through v_1, v_2 and v_3.
4. Let $A = \left\{ \begin{bmatrix} 2 \\ -1 \\ 5 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} -1 \\ 6 \\ 0 \end{bmatrix} \right\}$, $B = \left\{ \begin{bmatrix} 0 \\ 5 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ -3 \\ -2 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} \right\}$, and $\boldsymbol{n} = \begin{bmatrix} 3 \\ 1 \\ -2 \end{bmatrix}$. Find a hyperplane H with normal $\boldsymbol{n}$ that separates A and B. Is there any hyperplane that separates A and B strictly?

5. Let $A = \left\{ \begin{bmatrix} -2 \\ 1 \\ 5 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 5 \end{bmatrix}, \begin{bmatrix} -1 \\ 6 \\ 2 \end{bmatrix} \right\}$, $B = \left\{ \begin{bmatrix} 0 \\ 4 \\ -2 \end{bmatrix}, \begin{bmatrix} -1 \\ -3 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} \right\}$, and $\boldsymbol{n} = \begin{bmatrix} -3 \\ 2 \\ 2 \end{bmatrix}$. Find a hyperplane H with normal $\boldsymbol{n}$ that separates A and B. Is there any hyperplane that separates A and B strictly?